Kayla McFarlane

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EDUCATION

Carnegie Mellon University — **Pittsburgh, PA**
Master of Science in Robotics — *August 2027*
- Rales Fellow, Carnegie Mellon University (Aug 2025 – Aug 2027) — Full-tuition fellowship, **34 of 400+** applicants
- ETH Zurich Robotics Summer School (June 2025) — Selected as **1 of 50** students globally from **292** applicants

Bachelor of Science in Electrical and Computer Engineering (*Software Systems Concentration)* — *May 2025*

SKILLS

Programming: Python (NumPy, Pandas, Matplotlib), C++, C, Java, JavaScript, TypeScript, MATLAB, SQL
Tools & Frameworks: ROS2, Isaac Gym/Sim, Gazebo, Unreal Engine, PyTorch, React, Git, Linux, Docker, Conda
Technical Areas: Autonomous Robot Navigation, Risk-Aware Planning, Machine Learning, Reinforcement Learning, Simulation

RESEARCH EXPERIENCE

Graduate Research Assistant, CMU AirLab | SAFE Project — *May 2025 – Present*
- Designing a benchmark framework to evaluate adaptive safety behavior in navigation algorithms under stochastic uncertainty, using tail-risk metrics and outcome taxonomy to distinguish reckless from graceful behavior
- Creating safety-critical test scenarios with systematic risk event injection (dynamic obstacles with uncertain behaviors) to evaluate how algorithms perceive and respond to risk
- Implementing benchmark infrastructure in Isaac Sim/Gymnasium with Python, creating automated scenario generation and evaluation pipeline
- Authored workshop paper on multi-modal failure adaptation, accepted to IROS 2025 Workshop

Undergraduate Research Assistant, CMU AirLab | Safe Agile Function Everywhere Project — *Mar 2024 – May 2025*
- Conducted literature reviews on aerial robotics safety frameworks (collision avoidance, perception-aware navigation, reachability analysis) and evaluated state-of-the-art agile flight methods to identify research gaps
- Configured 3D simulation environments in Isaac Gym for testing perception-conditioned Neural Control Barrier Functions under degraded visual conditions
- Contributed to research integrating neural CBFs, Hamilton-Jacobi reachability, and Vision-Language Models for adaptive drone safety in cluttered environments

Undergraduate Research Assistant, CMU AirLab | Wildfire Project — *Jan 2023 – May 2025*
- Contributed to Wildfire Project, aimed at providing real-time situational awareness to firefighters during wildfires.
- Built automated data collection pipeline generating **15,000+** multimodal training images (RGB, thermal, depth)
- Created comprehensive testing framework to validate pipeline components, ensuring data quality for ML training
- Developed realistic smoke simulation in FIReVision using Unreal Engine for testing drone perception
- Presented research at CMU Meeting of the Minds Symposium and developed demo materials for Epic Games MegaGrant application

Study Bearbot – Multisensory HRI System | Capstone Project — *Jan 2025 – May 2025*
- Designed and evaluated multisensory desktop robot for student focus and stress reduction
- Conducted 8-participant user study demonstrating measurable stress reduction and increased focus
- Developed embedded firmware for Arduino Nano RP2040 with real-time sensor integration achieving **sub-500ms** responsiveness
- Built React-based interface for real-time robot control and study session tracking

Human-Robot Interaction Research Project — *Nov 2023 – Dec 2023*
Exploring Accent Variability, Phrase Difficulty, and Speaker Distance in Robot-Enabled Speech Recognition
- Evaluated Google Speech Recognition integrated with Pepper robot across diverse accents (American, Chinese, Guyanese, Brazilian) to assess ASR performance in classroom settings
- Developed Python modules for real-time transcription and analyzed performance using Translation Error Rate (TER) metrics
- Identified critical limitations in ASR accuracy for accented speech and distance variability, providing recommendations for improving accessibility in educational settings
- Project code adopted by AI Maker Space for demonstrating speech recognition challenges

INDUSTRY EXPERIENCE

Software Engineering Intern | DigITs Program | Memorial Sloan Kettering Cancer Center (MSK) — *Jun 2024 – Aug 2024*
- Developed AI chatbot interface using React and TypeScript for patient portal serving **170,000+** users annually
- Streamlined LLM fine-tuning process for **50+** developer teams, improving AI integration across departments
- Led development of MSK SkinMonitor, **winning 1st place** in intern hackathon for AI-powered health tracking app
- Practiced Agile methodologies through daily stand-ups and two-week sprint cycles

PUBLICATIONS

- Navin Sriram Ravie, Andrew Jong, **Kayla McFarlane**, Jay Patrikar, Bijo Sebastian, Sebastian Scherer. "Don't Fool Me Twice: Adapting to Failures in the Wild with Multi-Modal Reasoning." *IROS 2025 Workshop on The Art of Robustness: Surviving Failures in Robotics*, October 2025.